THE BMOA CONTACT-SET CONVERSE

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AI assistance was used in adapting and checking the argument and the exposition.

Laitila, Nieminen, Saksman, and Tylli ask in Problem 4.3(3): "Is the converse of Proposition 4.1(i) also true?" The proposition states that ℓ^2-singularity of C_φ on BMOA implies $m(E_\varphi) = 0$, where $E_\varphi = \{\zeta \in \mathbb{T} : |\varphi(\zeta)| = 1\}$.

Proposition 4.1(i) says that if C_φ is ℓ^2-singular on BMOA, meaning that it is not bounded below on any subspace isomorphic to ℓ^2, then $m(E_\varphi) = 0$; m is normalized arc length.

Laitila, Nieminen, Saksman, and Tylli, *Rigidity of composition operators on the Hardy space H^p*, arXiv:1607.00113.

Answered: no. Evidence: Sketch.

No.

Proof. Let $a_j = 1 - 2^{-j^2}$, let B be the Blaschke product with these zeros, and set $\varphi(z) = (1+z)B(z)/2$. Since B is inner, $|\varphi(\zeta)| = |1+\zeta|/2 < 1$ almost everywhere except at $\zeta = 1$, so $m(E_\varphi) = 0$.

The zeros are thin. For $0 < s < t < 1$, their pseudohyperbolic distance satisfies

$$1 - \rho(s,t) = \frac{(1-t)(1+s)}{1-st} \le \frac{2(1-t)}{1-s}.$$

Thus for $k \neq j$, $1 - \rho(a_j, a_k) \le 2\,2^{-|j^2-k^2|} \le 2\,2^{-j|j-k|}$, and consequently

$$\sum_{k\neq j}(1 - \rho(a_j,a_k)) \le 4\sum_{m\ge 1} 2^{-jm} \le 8\,2^{-j} \longrightarrow 0.$$

It follows that $\delta_j = \prod_{k\neq j}\rho(a_j,a_k) \to 1$. Write $B = \sigma_{a_j}B_j$. Then $\varphi\circ\sigma_{a_j}(z) = zh_j(z)$ with $|h_j(0)| = (1+a_j)\delta_j/2 \to 1$. Schwarz's lemma gives $|h_j| \le 1$; by normality a subsequence converges locally uniformly to a unimodular constant λ. Hence $\varphi\circ\sigma_{a_j} \to \lambda z$ along that subsequence.

For $f \in$ BMOA, the seminorm is $\sup_{a\in\mathbb{D}}\|f\circ\sigma_a - f(a)\|_{H^2}$. Since $\varphi(a_j) = 0$, the corresponding tests for $C_\varphi f$ converge locally uniformly to $f(\lambda z) - f(0)$. Each test norm is at most $\|C_\varphi f\|_{\mathrm{BMOA}}$ by the defining seminorm, and lower semicontinuity of radial H^2 means gives

$$\|C_\varphi f\|_{\mathrm{BMOA}} \ge \|f(\lambda\cdot) - f(0)\|_{H^2} = \|f - f(0)\|_{H^2}.$$

Paley's lacunary theorem identifies the closed span of z^{2^k}, $k \ge 1$, with ℓ^2 in BMOA. The displayed inequality makes C_φ bounded below on this copy, so it is not ℓ^2-singular. This disproves the converse. □

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